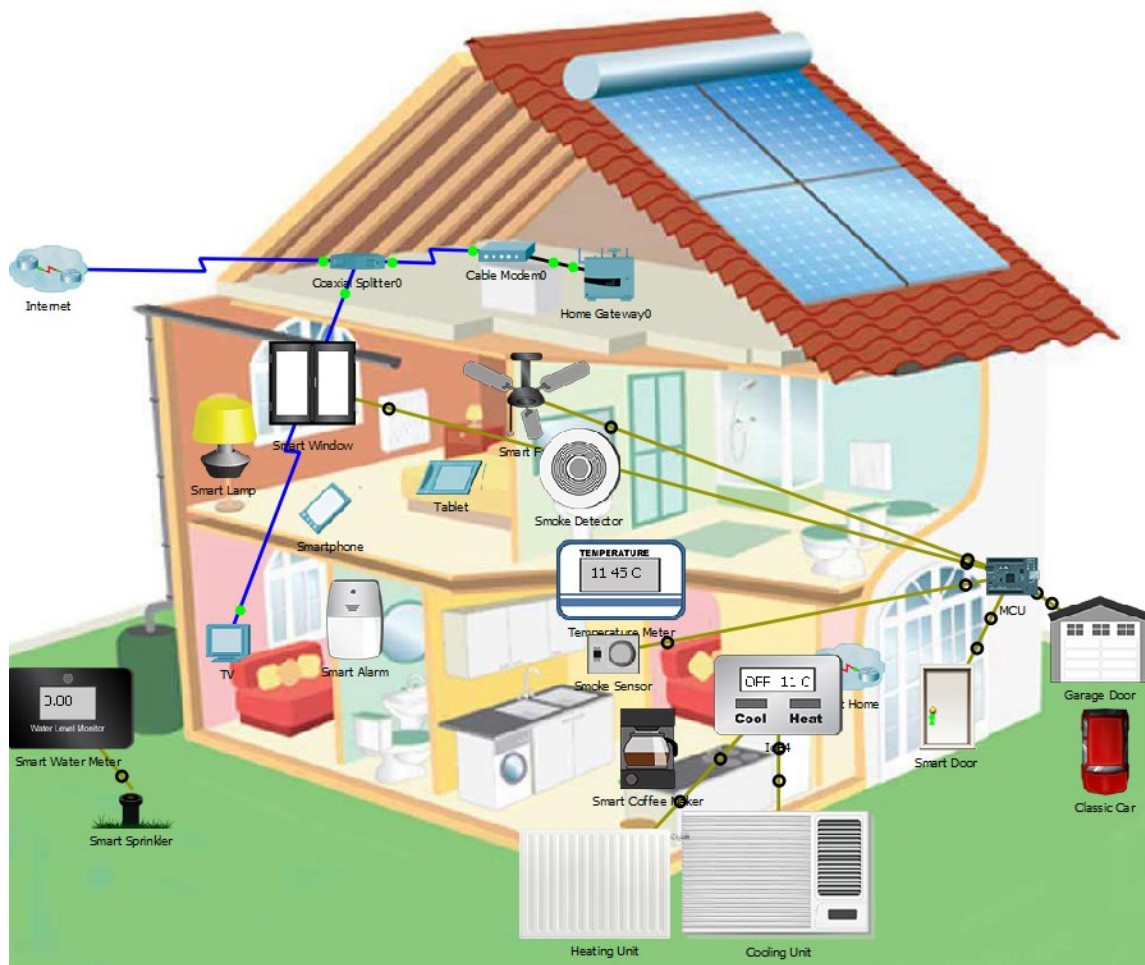


Packet Tracer – Threat Modeling at the IoT Communication Layer

Topology

The topology is a home IoT system that has been prototyped in Packet Tracer. It shows a cutaway view of a home with different sensors, actuators, and connections shown.



Objectives

In this Packet Tracer, you will begin the threat modeling process for the device layer of the IoT attack surface.

Part 1: Diagramming the Communication Layer

Part 2: Creating an Inventory of Assets of the Communication Layer Attack Surface

Part 3: Identifying Potential Threats with the STRIDE Model

Background / Scenario

In this Packet Tracer, you will continue creating a threat model for the home automation IoT system that was started in the **Chapter 3 Packet Tracer - Threat Modeling at the IoT Device Layer**. You may want to review the description of the system and Packet Tracer file from that Packet Tracer.

You will be working with the communication layer of the IoT attack surface. Remember that the home automation network differs somewhat from the PT network. The home automation network that you are working with does not use Wi-Fi as the wireless protocol for the sensors and actuators. Instead, you will choose another protocol for the IoT sensor-actuator-controller network, such as Zigbee or Z-Wave. The controller handles local communication with the various wireless networks in the house and the Internet. It serves as the IP gateway for the IoT sensor-actuator network. It also functions as a router that enables communication between the local wireless LAN and the Internet.

Required Resources

- Packet Tracer 7.1 or later

Part 1: Diagramming the Communication Layer

Step 1: Choose a protocol for the sensor network.

Focus on the IoT communication layer protocols and infrastructure. Although the PT model uses Wi-Fi for communication between sensors, actuators and the IoT gateway, you will use another protocol for your analysis.

Choose a protocol that is commonly used in home automation networks for communication between sensors, actuators and the home IoT gateway. Keep security in mind.

What protocol did you choose and why did you choose it?

Step 2: Diagram the network.

Use your completed floorplan diagrams from the Threat Modeling at the IoT Device Layer Packet Tracer from Chapter 3. Additionally, you will complete the attic floorplan.

- Complete the floorplan for the attic that is included in **Appendix A** of the PDF version of these Packet Tracer instructions. Add the network infrastructure devices and their connections as they are shown in the Packet Tracer network.
- Add the network clients that are not IoT devices but are still connect to the IoT gateway.
- Add data flows. **It is not necessary to draw flows for every device**, just a few representative devices is fine. In addition, when indicating connections between floors, it is fine to just write "Connected to..." at the top of a floorplan. Include several sensors and actuators, the smart phone and tablet, and the MCU.

Draw lines between the IoT devices, the IoT gateway, and the other network infrastructure components which are connected to one another. Use different colors for connections that use different protocols. Remember that some sensors and actuators may not be able to communicate directly with the IoT gateway due to power constraints. Label or color code the data flows according to the protocols in use.

What would allow any sensors and actuators that are out of range of the IoT gateway to communicate with it?

What other protocols are in use in the network? How could the tablet and the smart phone communicate?

- d. Add arrows to indicate the predominate direction of data flow. In some cases the flows will be mostly in one direction. In others they will be in two directions.

Which devices in the home automation network connect to the IoT Gateway?

Part 2: Creating an Inventory of Assets of the Communication Attack Surface

Another component of asset management is understanding the network protocols and infrastructure devices that are in use on the network. In this part of the Packet Tracer, you will complete a table for the all of the communication attack surface assets.

Step 1: List the communication networks and protocols.

Open the Packet Tracer activity and fill-in the table that is provided below. The table should include all the networks, protocols, and IP devices that are part of the home automation network system. Determine the communication relationships between assets. The network is a collection of things that uses the same protocol. It is not necessary to list every IoT device here. Instead, just refer to the wireless sensor-actuator network. This has been filled in for you.

Follow the instructions in the Packet Tracer activity to show the wireless network connections between devices if necessary. Feel free to move devices around to understand the wireless connections better. Pay attention to the color coding of the wireless connections.

Network or Device	Protocol(s)	Communicates With
Part 2: Communications		

Part 3: Identifying Potential Threats

In this part, you will identify threats using the STRIDE methodology. Try to describe as many threats as possible based on your experience in the course, the [OWASP IoT vulnerabilities page](#), and other information sources.

Use the STRIDE model to create a list of potential threats.

Fill in the following table with threats for each category in the STRIDE threat model. Add potential threats that could occur for each STRIDE category. Include the type of threat using the OWASP terminology where possible. An extra STRIDE table is provided in Appendix B of the PDF version of these Packet Tracer instructions.

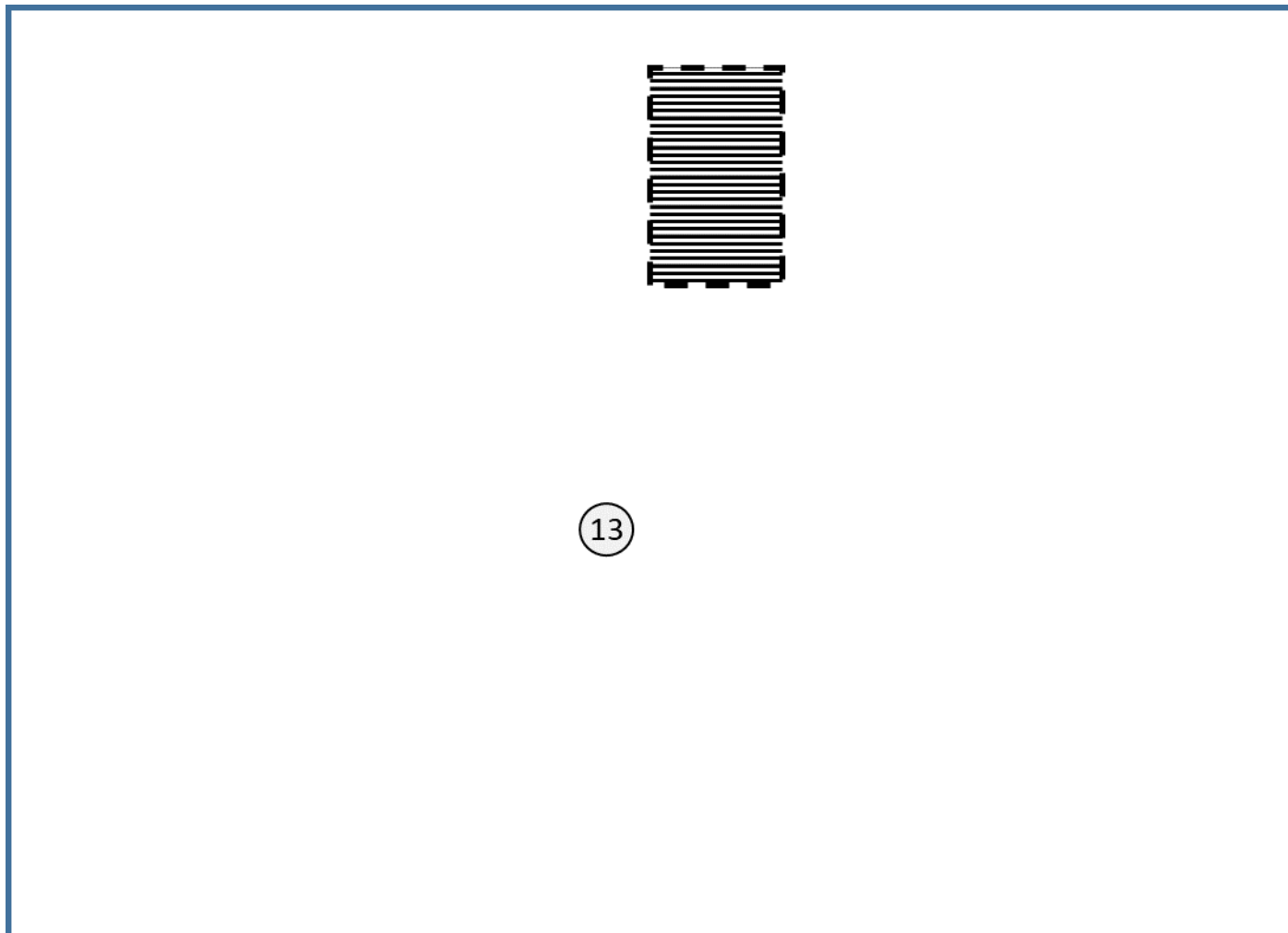
Threat type	Network or Device	Threats
(S)poofing – can an attacker pretend to be someone he's not, or falsify data?	sensor-actuator network	

Threat type	Network or Device	Threats
(T)ampering – can an attacker successfully inject falsified data into the system?	sensor-actuator network	
(R)epudiation – can a user pretend that a transaction didn't happen?	sensor-actuator network	
(D)enial of Service – can the device be shut down or made unavailable maliciously?	sensor-actuator network	

Threat type	Network or Device	Threats
(E)scalation of Privilege – can users get access to privileged resources meant only for admins or superusers?	sensor-actuator network	

Appendix A: Floor Plans

Attic:



Ground Floor Legend:

Number	Room
13	attic

Appendix B: STRIDE Table

Threat type	Network or Device	Threats
(S)poofing – can an attacker pretend to be someone he's not or falsify data	sensor-actuator network	
(T)ampering – can an attacker successfully inject tampered data into the system?	sensor-actuator network	

Threat type	Network or Device	Threats
(R)epudiation – can a user pretend that a transaction didn't happen?	sensor-actuator network	
(D)enial of Service – can the device be shut down or made unavailable maliciously?	sensor-actuator network	

Threat type	Network or Device	Threats
(E)scalation of Privilege – can users get access to privileged resources meant only for admins or superusers?	sensor-actuator network	